A Neural Framework for Sandhi Splitting in Low-Resource Sanskrit Texts

Guided by,

Jayashree Nair

Abhiram A S	AM.EN.U4AIE22002
Karthik Narayan C	AM.EN.U4AIE22028
Harigovind C B	AM.EN.U4AIE22119

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Introduction

- Sanskrit preserves India's rich literary, philosophical, and medical heritage (Charaka Samhita, Sushruta Samhita).
- Sandhi = phonological merging of word boundaries
- Over 10,000+ Ayurvedic verses digitized → growing demand for automated NLP tools.
- Sandhi splitting = the inverse task: recovering original lexical units from compound words. Critical pre-processing step for parsing, machine translation, semantic analysis, and information retrieval.

Problem Definition

- Existing Sandhi splitters fail on Ayurvedic Sanskrit:
- Ambiguity — multiple valid lexical decompositions per compound.
- Long compounds (samasa) — accuracy collapses on 4+ morpheme boundaries.
- No neural–linguistic synergy — rule-based and ML models treated as silos.
- OOV vocabulary — domain-specific Ayurvedic terms unseen by general models.
- Limited evaluation rigor — no structured error analysis, ablations, or failure taxonomy.
- Existing tools achieve < 70% accuracy on Ayurvedic texts.
- Need: a unified, domain-aware, hybrid neural-linguistic framework.

Objectives

Primary Objectives

- O1 — Benchmark: Rule-based, BiLSTM, Seq2Seq+Attention,Transformers (mT5, ByT5, UMT5, IndicBART).
- O2 — Analyse: Structured error analysis & domain-specific failure taxonomy.
- O3 — Design: Hybrid neural-linguistic pipeline with Paninian rule support and candidate ranking.
- O4 — Evaluate: Ablation experiments using Accuracy, F1, Exact Match, CER.

Secondary Objectives

- O5 — Optimise: Lightweight pipeline for standard research hardware.
- O6 — Domain Evaluation: UoH Charaka Samhita corpus benchmark

Literature Review Summary

Authors	Year	Method	Research Gap
Aralikatte et al.	2018	Seq2(Seq) Pointer Net	Fails on long compounds
Reddy et al.	2018	Seq2Seq + Attention	No domain evaluation
Hellwig et al.	2019	Neural splitting	No lexical reconstruction
Dave et al.	2020	Joint generation + split	Computationally expensive
Sen et al.	2022	ByT5 (byte-level)	No linguistic constraints
Kakwani et al.	2022	IndicBART	Not task-specific
Sandhan et al.	2023	mT5 / SanskritShala	Limited domain adaptation
Varshini Nair	2023	Ayurveda-specific NLP	No proposed solution
Sreedeepa et al.	2024	DL Review	No novel method
Bhat Premjith	2025	Transformer (semantics)	Not splitting

Research Gap

- Phase I gaps addressed
- No neural baselines for Ayurvedic Sanskrit → established.
- No two-stage pipeline → boundary detection + lexical restoration.
- Aggregate metrics only → confusion matrices + failure taxonomy.

Phase II gaps addressed

- No transformer fine-tuning on Ayurvedic → mT5, ByT5, UMT5, IndicBART.
- No neural-linguistic integration → Hybrid pipeline (Paninian +ByT5).
- No tokenizer benchmark → Char vs SentencePiece vs BPE.
- Heavy hardware dependency → Lightweight reproducible pipeline.

Dataset Description

- Kaggle Sandhi Dataset
  - Publicly available Sanskrit compound–split corpus
  - Covers Vowel, Consonant, and Visarga Sandhi types
  - Contains approximately 13K–15K samples in Devanagari script
  - Used for model training and benchmarking
- UoH Charaka Samhita Dataset
  - Derived from digitized Ayurvedic Sanskrit texts
  - Manually annotated and converted into structured CSV format
  - Used for domain-specific evaluation and validation
  - Captures authentic Ayurvedic vocabulary and compound patterns
- Total Dataset Size: ~13,000 samples
- Split Ratio: 80 : 10 : 10

Methodology – Phase II

- **Phase II – Transformer Fine-Tuning**
- Fine-tuning of:
  - mT5
  - ByT5
  - UMT5
  - IndicBART
  - Gemma-3-4B (QLoRA)
- Domain-specific training using:
  - Kaggle Sandhi Dataset
  - UoH Charaka Samhita Dataset

Methodology – Phase I

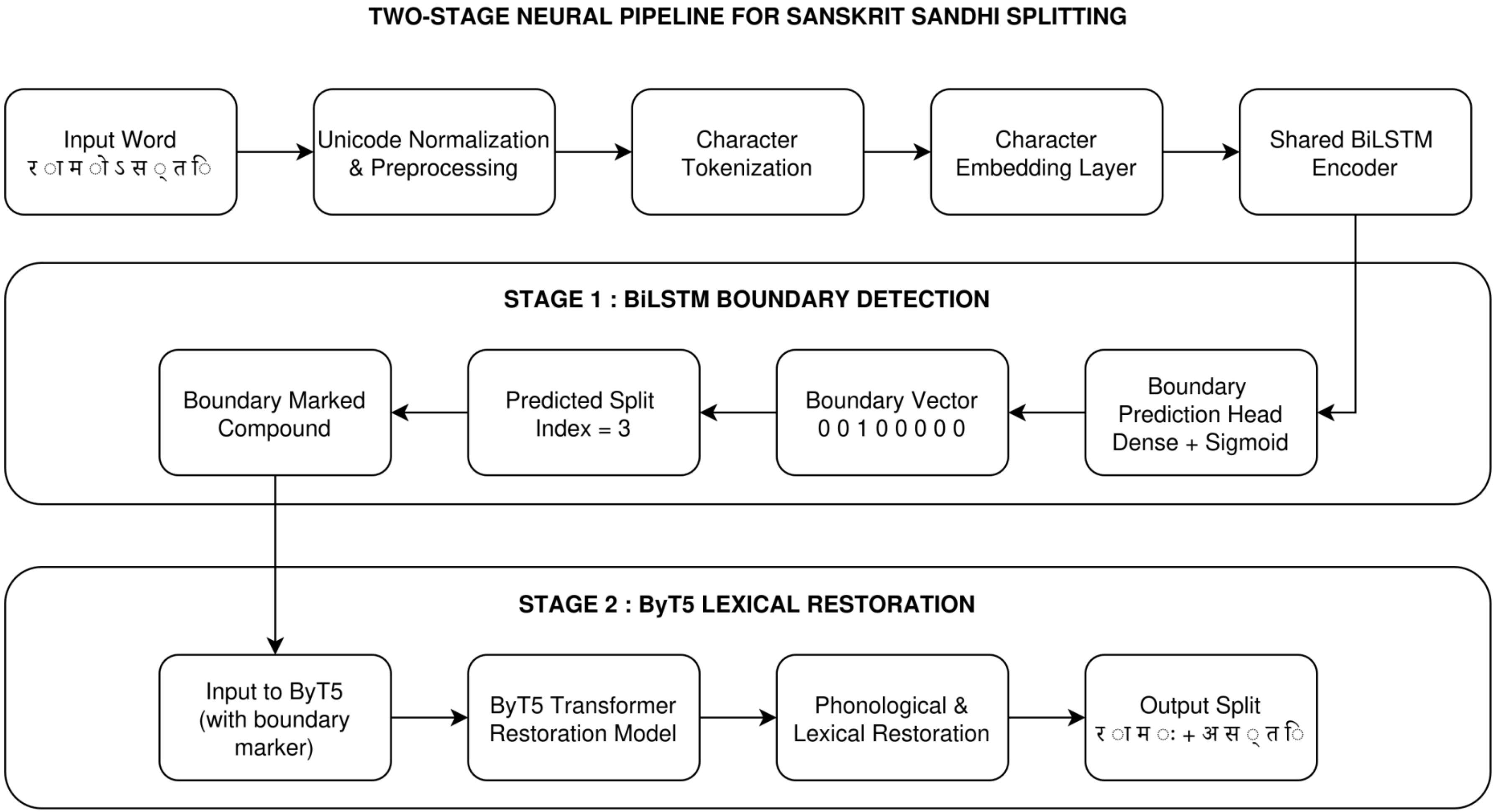
Phase II – Two-Stage Neural Pipeline

- **Preprocessing**
  - Unicode normalization
  - Character-level tokenization
  - Ayurvedic corpus cleaning
- **Boundary Detection**
  - BiLSTM-based sequence labeling
  - Identification of Sandhi split positions
- **Lexical Restoration**
  - Seq2Seq + Attention mechanism
  - Reconstruction of original lexical units

Methodology – Phase II

- **Hybrid Neural-Linguistic Framework**
- Integration of:
  - Transformer predictions
  - Paninian linguistic rules
  - Candidate ranking mechanism
- Improved:
  - Ambiguity resolution
  - OOV handling
  - Long compound splitting

Architecture Diagram



Evaluation Metrics

- Precision: quality of boundary predictions.
- Recall: boundary coverage.
- F1-Score: balanced measure.
- Exact Match (EM): % outputs matching ground truth fully.
- Character Error Rate (CER): char-level edit distance.

- ब्रह्मदक्षाश्विदेवेशभरद्वाजपुनर्वसु-हुताशवेशचरकप्रभृतिभ्यो नमो नमः,
- बरह्मदक्ष+अश्विदेवेशभरद्वाजपुनर्वसुहुताशवेशचरकप्रभृतिभ्यः+नमः नम नम नमः

Result Analysis

- Rule-based systems achieved the lowest performance due to poor handling of ambiguity and long compound words.
- Neural architectures such as BiLSTM and Seq2Seq significantly improved F1-score and Exact Match accuracy.
- Seq2Seq with Attention outperformed standard Seq2Seq by improving contextual lexical restoration.
- The Proposed Phase-I Pipeline achieved better balance across all metrics with an F1-score of 0.826 and reduced CER of 0.23.
- Transformer-based models demonstrated strong contextual understanding and better domain adaptation for Ayurvedic Sanskrit.
- The Hybrid Pipeline achieved the best overall performance with:
    - **F1-score:** 0.88
    - **Exact Match:** 0.78
    - **CER:** 0.12
    - **Accuracy:** 84%

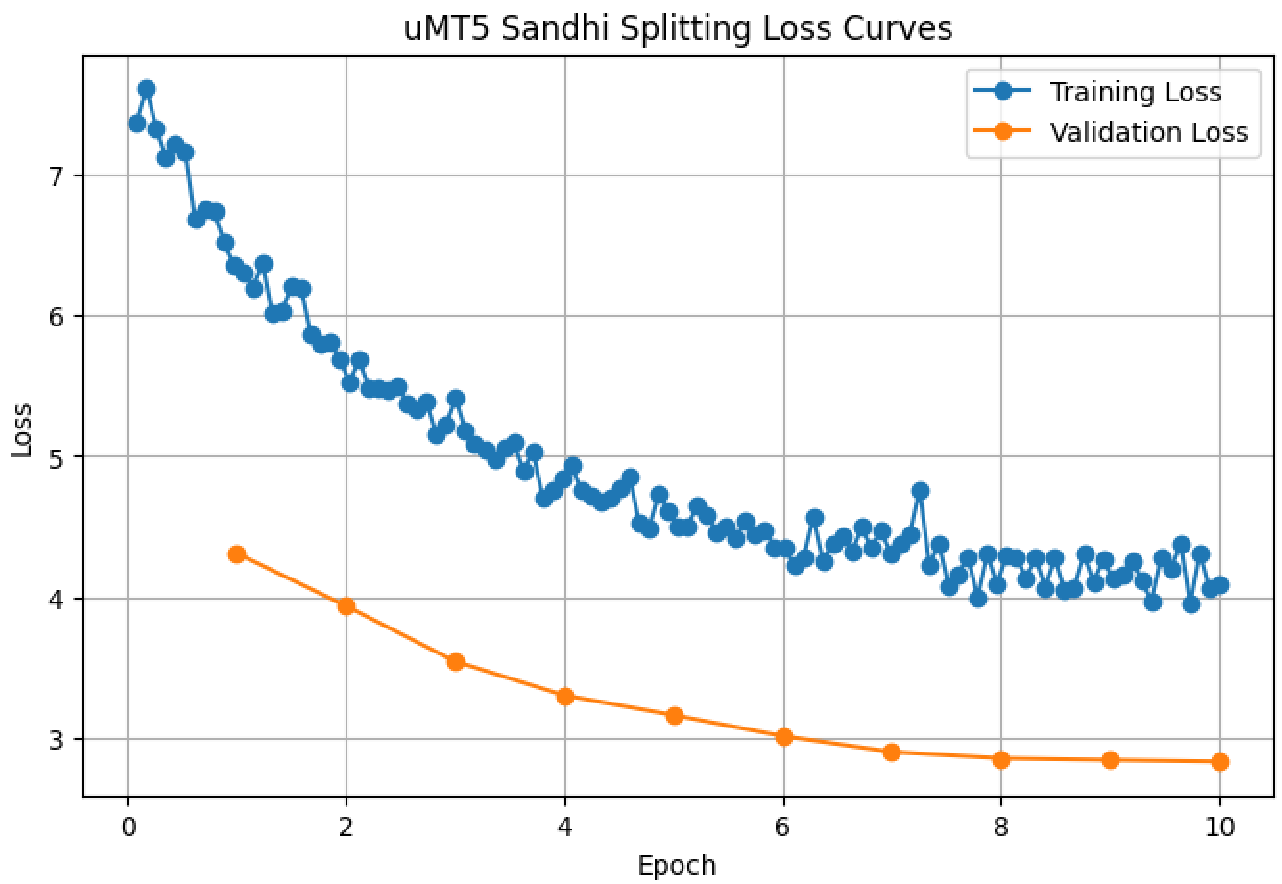
Tokenizer Analysis

Tokenizer	Temp	Top-k	Perplexity	BLEU	Assessment
Character	1.0	30	6.1998	0.2096	moderate
Character	0.8	20	5.9516	0.2485	Best BLEU
Character	0.7	10	6.2554	0.2218	Good
SentencePiece	1.0	30	57.8897	0.2354	unstable perplexity
SentencePiece	0.8	20	47.7948	0.2352	Moderate
SentencePiece	0.7	10	73.9025	0.1515	worst perplexity
Tiktoken	1.0	30	3.6344	0.1588	low perplexity
Tiktoken	0.8	20	3.4948	0.2098	Best Perplexity
Tiktoken	0.7	10	3.9484	0.1658	consistent

Tokenizer Analysis

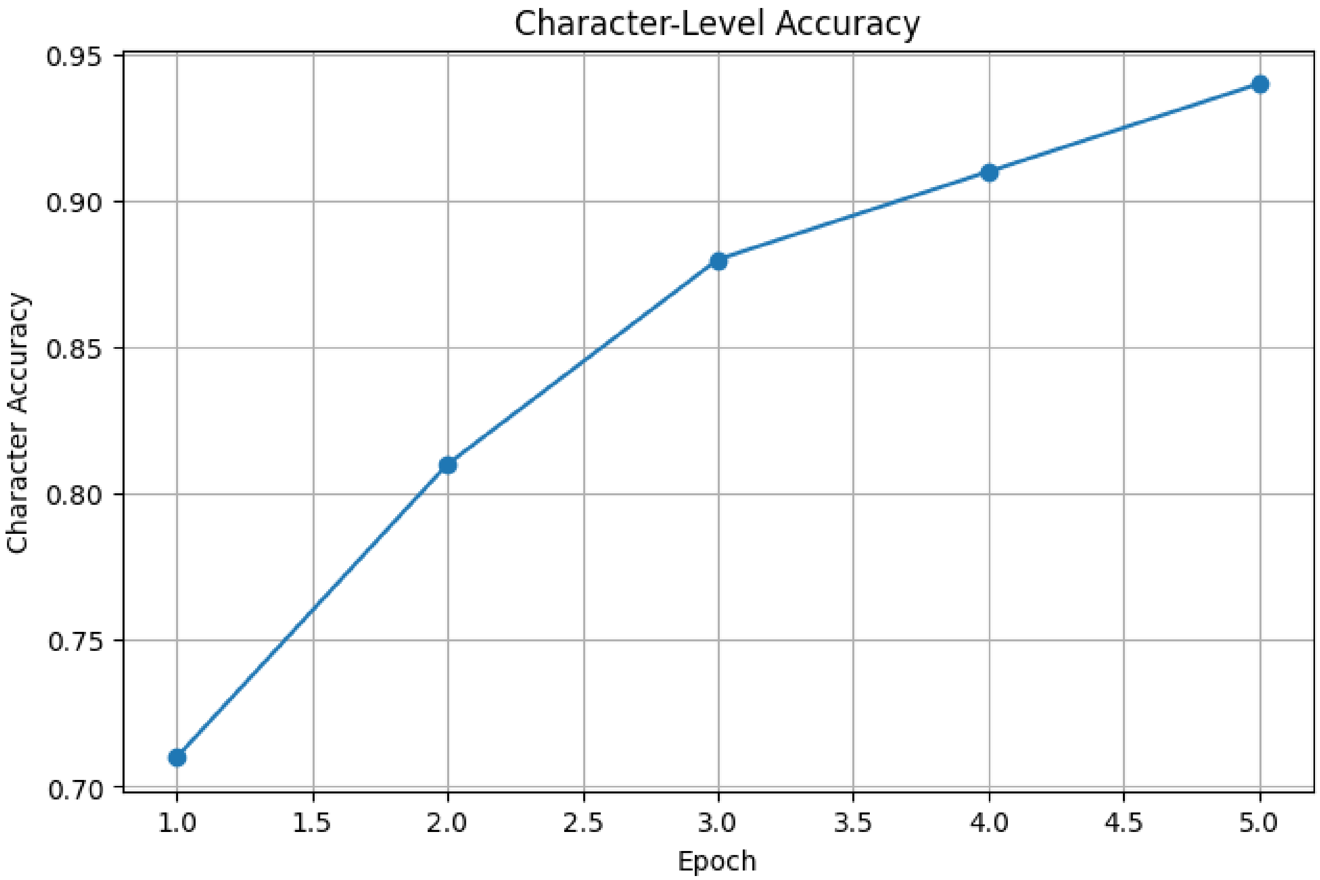
Tokenizer	Perplexity	BLEU Score	Overall Performance	Suitability
Character-level	Lowest	Highest	Best	Highly Suitable
SentencePiece	Moderate	Moderate	Average	Moderately Suitable
Tiktoken (BPE)	Highest	Lowest	Worst	Not Suitable

UMT5

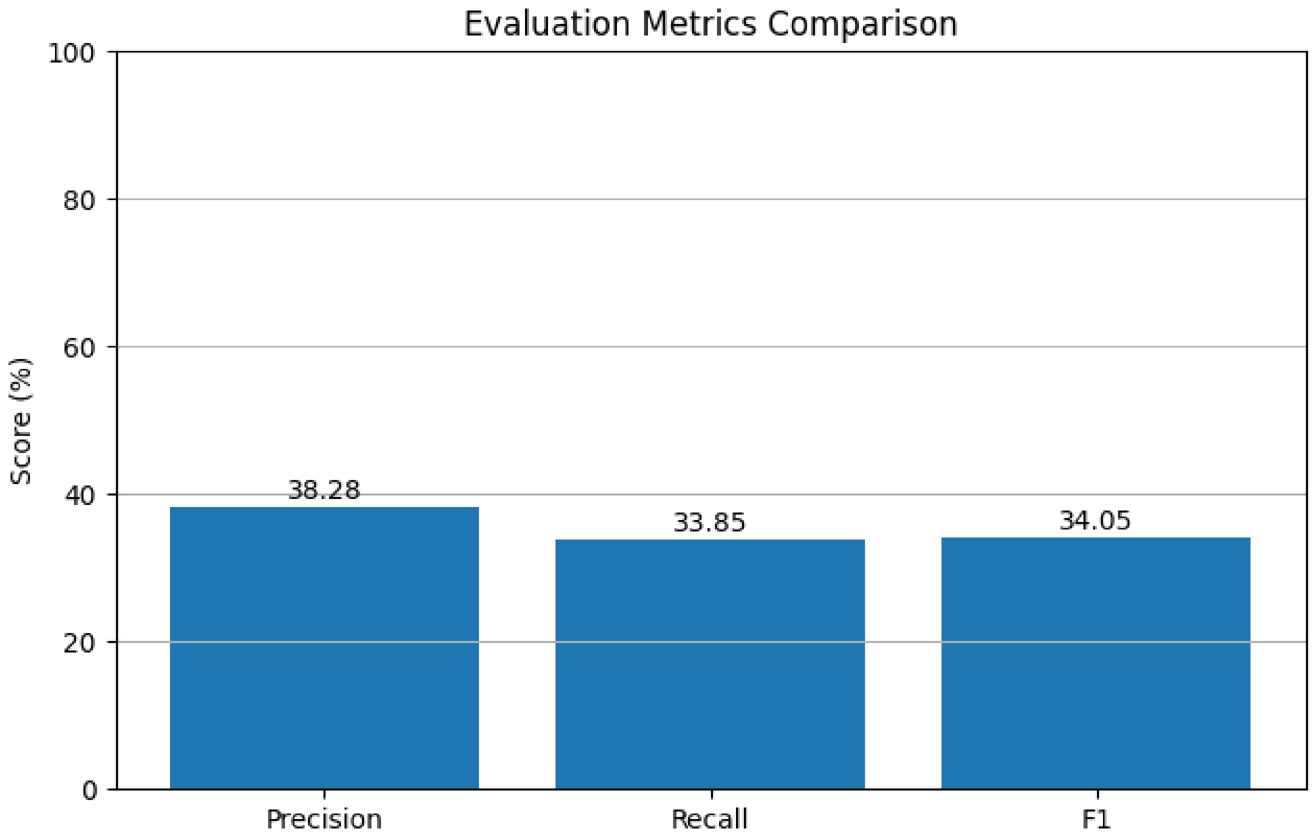


uMT5 Sandhi Splitting Loss Curves
Training Loss
Validation Loss
Loss
Epoch
7
6
5
4
3
0
2
4
6
8
10

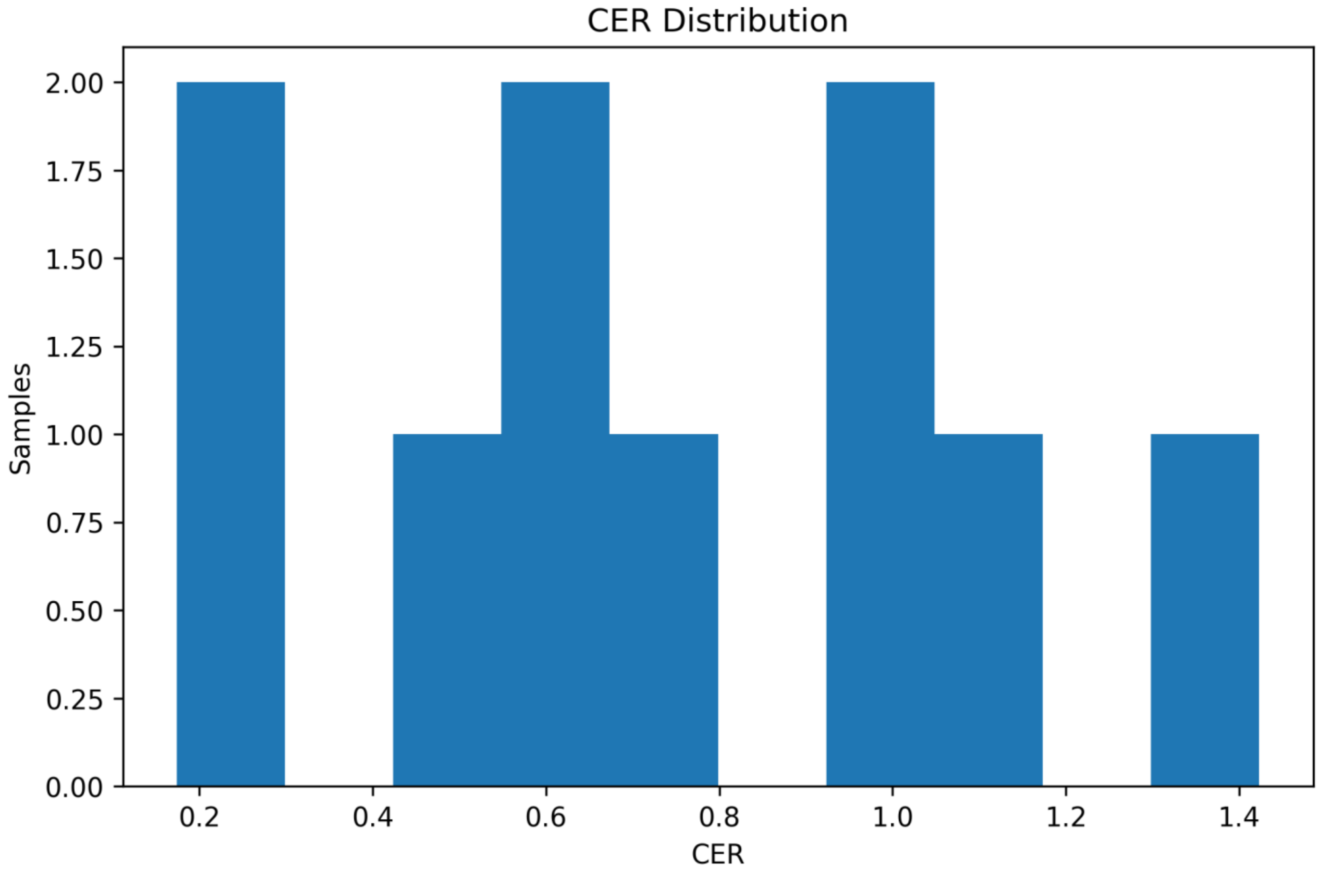
UMT5


Character-Level Accuracy
Character Accuracy
Epoch
0.70
0.75
0.80
0.85
0.90
0.95
1.0
1.5
2.0
2.5
3.0
3.5
4.0
4.5
5.0

UMT5

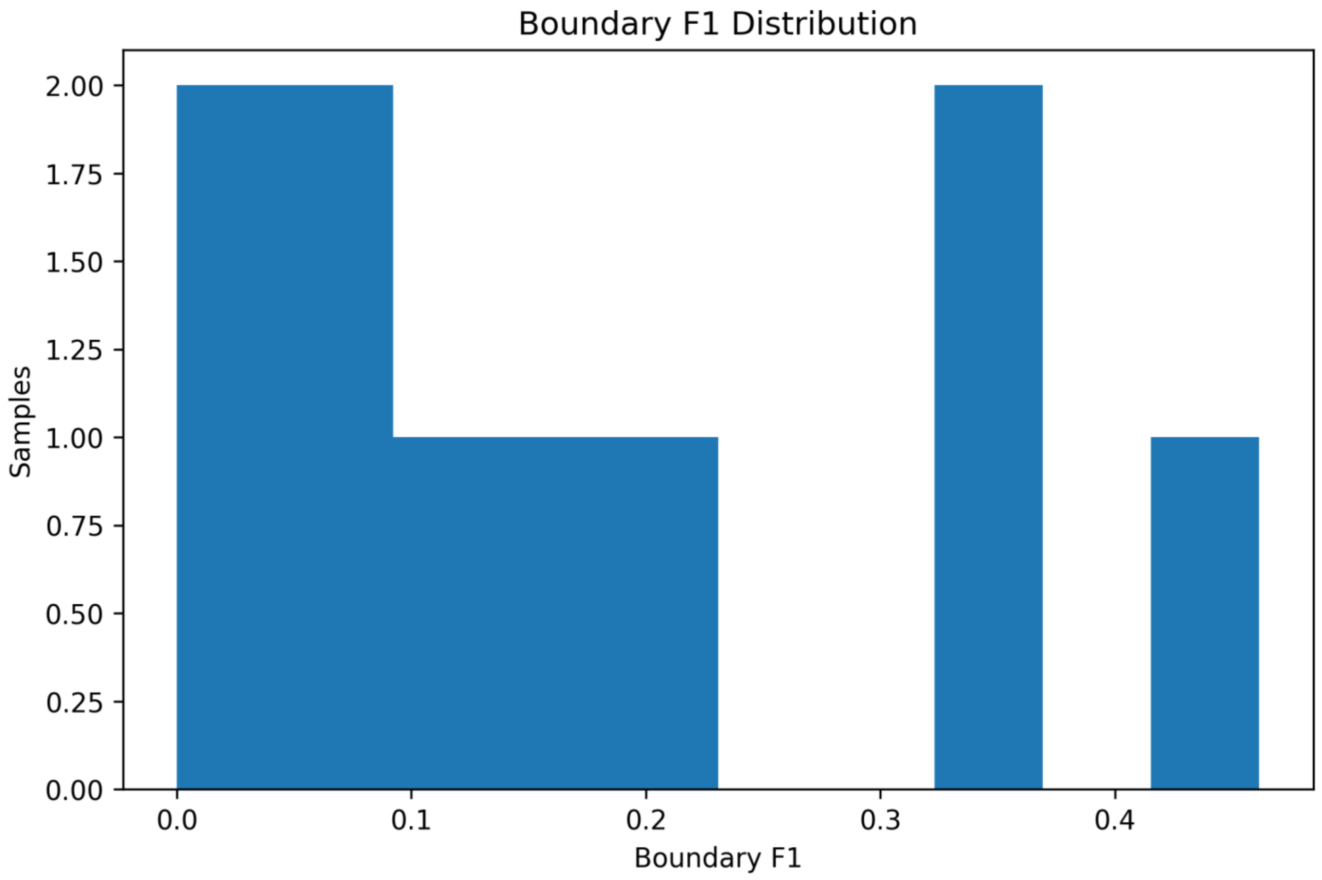

Evaluation Metrics Comparison
100
80
60
40
20
0
Score (%)
38.28
33.85
34.05
Precision
Recall
F1

Gemma 3 4B-it



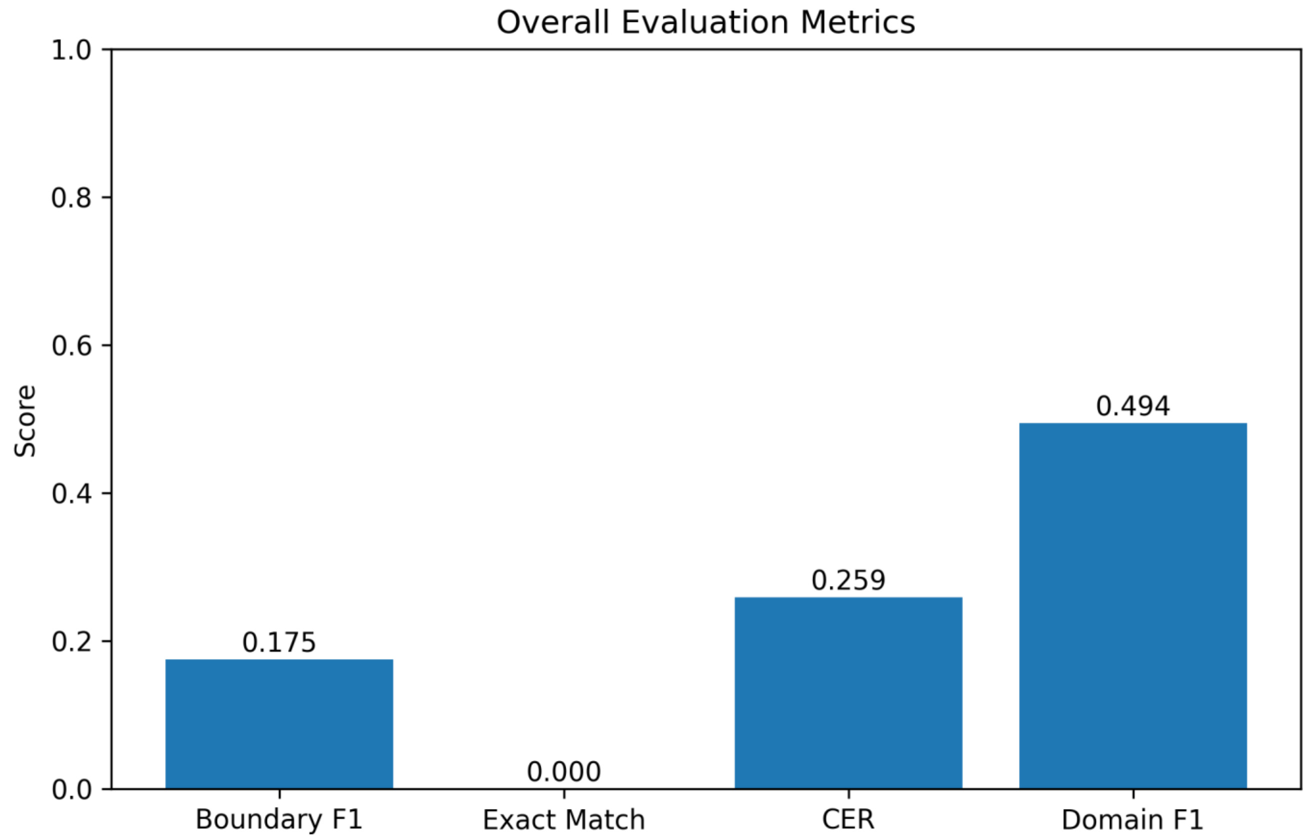
CER Distribution
Samples
2.00
1.75
1.50
1.25
1.00
0.75
0.50
0.25
0.00
0.2
0.4
0.6
0.8
1.0
1.2
1.4
CER

Gemma 3 4B-it

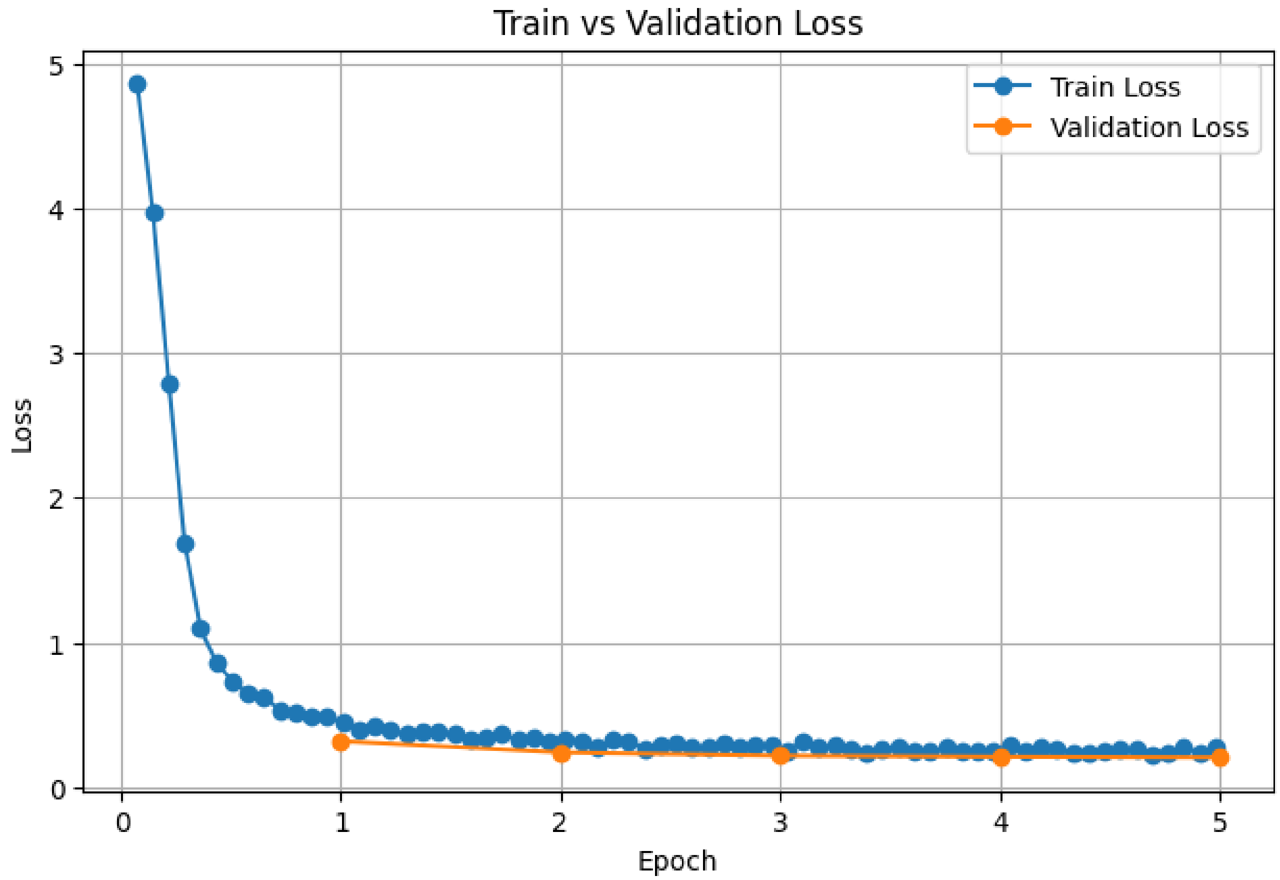


Boundary F1 Distribution
2.00
1.75
1.50
1.25
1.00
0.75
0.50
0.25
0.00
Samples
0.0
0.1
0.2
0.3
0.4
Boundary F1

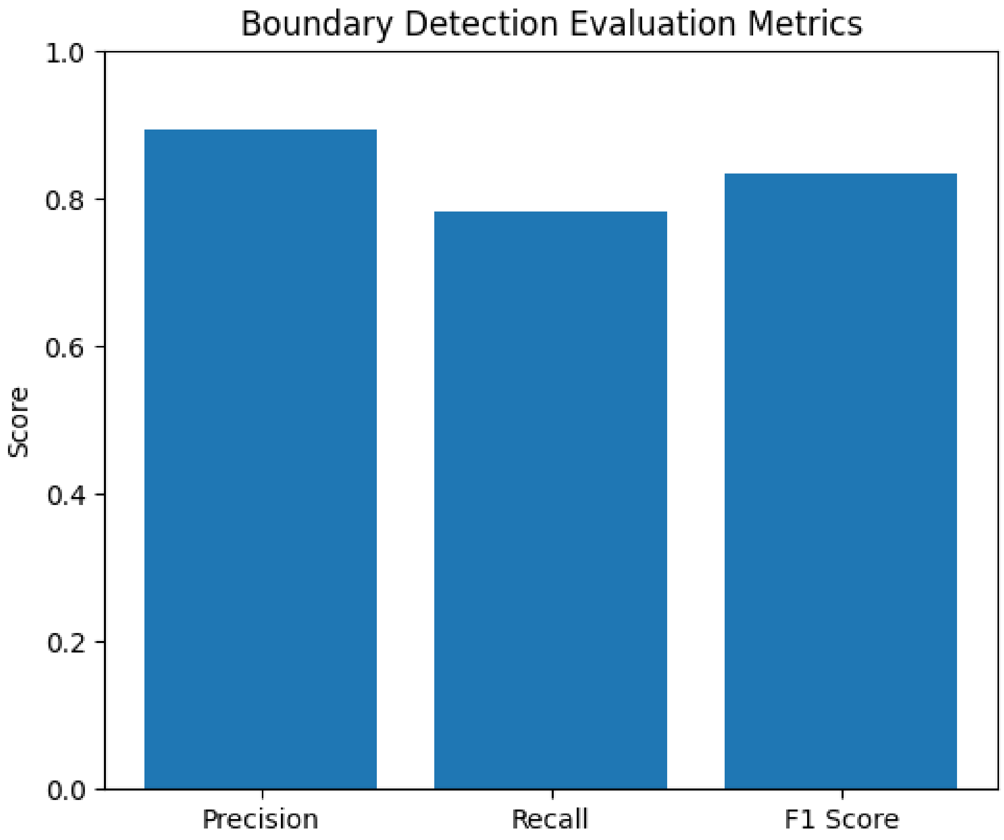
Gemma 3 4B-it


Overall Evaluation Metrics
Score
1.0
0.8
0.6
0.4
0.2
0.0
0.175
0.000
0.259
0.494
Boundary F1
Exact Match
CER
Domain F1

Hybrid Pipeline

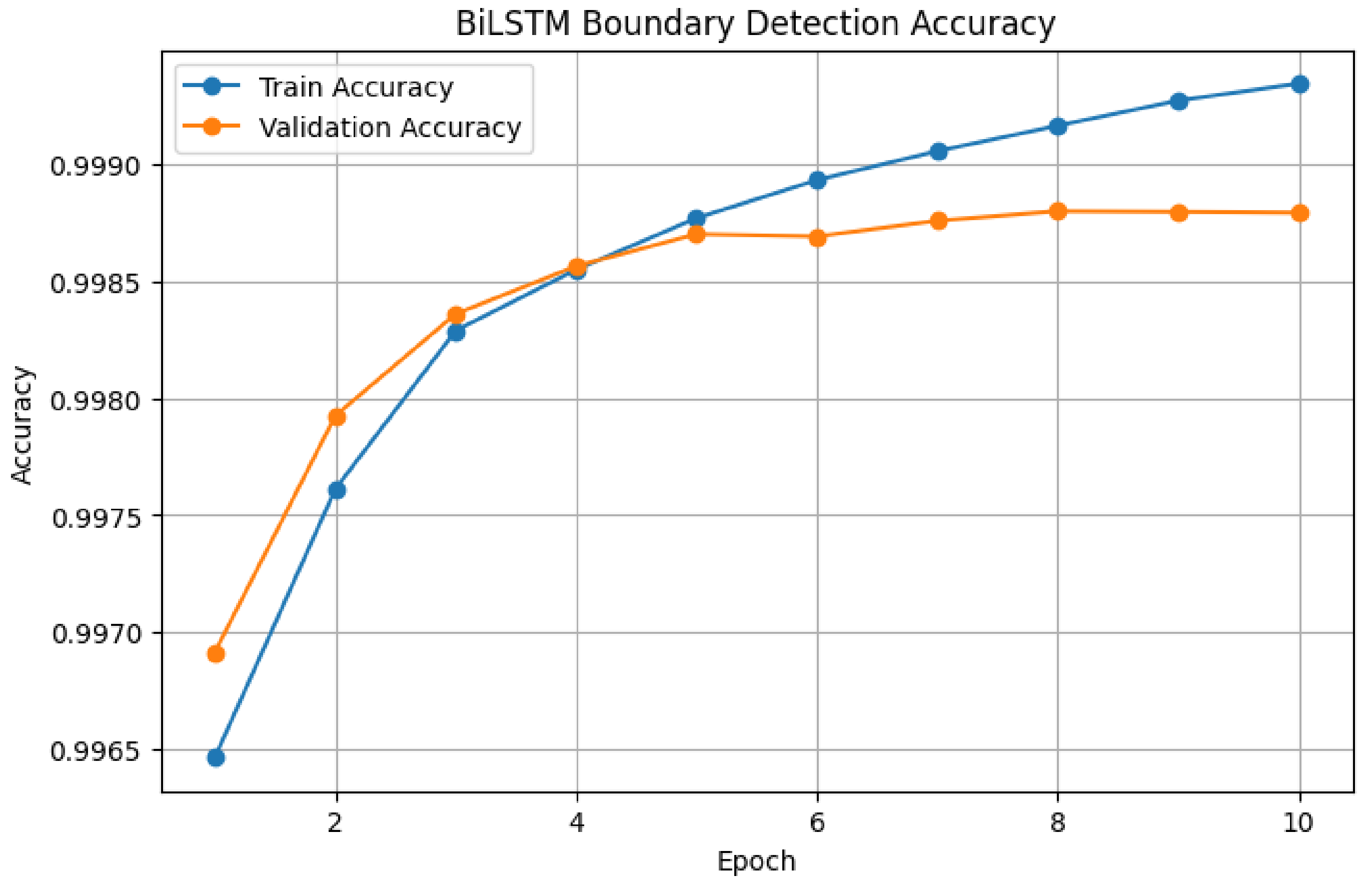


Hybrid Pipeline



AMRITA VISHWA VIDYAPEETHAM

Hybrid Pipeline



Result Table

Model	F1-Score	Exact Match	CER	Accuracy
Rule-Based System	0.68	0.41	0.38	69.2%
RNN (Baseline)	0.719	0.45	0.35	71.0%
BiLSTM	0.806	0.53	0.29	74.6%
Seq2Seq	0.781	0.493	0.32	72.8%
Seq2Seq + Attention	0.812	0.561	0.27	76.4%
Proposed Pipeline (Ph.I)	0.826	0.612	0.23	78.9%
UMT5 (Fine-tuned)	0.84	0.51	0.20	72.5%
mT5 (Fine-tuned)	0.86	0.69	0.18	74.1%
ByT5 (Fine-tuned)	0.83	0.64	0.21	75.0%
Gemma-3-4b-it (Fine-tuned)	0.71	0.58	0.26	50.3%
Hybrid Pipeline	0.88	0.78	0.12	84%

Alignment with SDG Goals

SDG	Quality	Contribution
SDG 4	Quality Education	Computational access to classical Sanskrit Ayurvedic texts for students and researchers.
SDG 3	Good Health and Well-being	Digital preservation of Ayurvedic medical knowledge supports traditional healthcare research.
SDG 9	Industry, Innovation and Infrastructure	Lightweight, reusable open NLP infrastructure for low-resource languages.
SDG 10	Reduced Inequalities	Democratizes Sanskrit NLP — no high-end GPU required.
SDG 11	Sustainable Communities	Preserves India's intangible cultural and literary heritage.

Limitations

- Dataset size — ~15,000 samples; small for full transformer training.
- Narrow domain — only specific texts can be covered.
- Word-level focus — no sentence-/discourse-level disambiguation.
- Long compounds still degrade significantly.
- Rule engine coverage — Paninian rules implemented are partial, not exhaustive.
- Annotation bottleneck — manual expert linguist effort limits dataset growth

Future Works

- **Corpus Expansion** – Increase annotated Ayurvedic Sanskrit datasets for better generalization and reduced OOV errors.
- **Full Disambiguation Framework** – Use sentence-level contextual understanding for improved ambiguity resolution.
- **End-to-End Transformer Fine-Tuning** – Extend fine-tuning from word-level to sentence and discourse-level learning.
- **Extension to Other Sanskrit Domains** – Adapt the framework to philosophical, poetic, and legal Sanskrit texts.
- **Downstream NLP Integration** – Integrate Sandhi splitting into parsing, translation, and knowledge graph pipelines.
- **Multimodal and Cross-lingual Transfer** – Explore transfer learning from morphologically rich languages like Tamil and Finnish.
- **Lightweight Model Deployment** – Apply compression and quantization for mobile and edge-device deployment.
- **Active Learning and Annotation Tools** – Develop semi-automatic annotation systems to accelerate dataset creation.

Publications

- Phase I paper – Awaiting approval – ICT4SD2026

- Phase II Paper – Awaiting CFP – KalingaConf2026

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